Memory transfer and continual learning in predictive-coding associative networks

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Abstract

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Author summary

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1 Introduction

2 Model and analysis

2.1 A two-population predictive-coding memory

2.2 Student inference and learning descend a free energy

Proposition 1. *[Student state relaxation is exact gradient descent on F_S; the zero-diagonal weight update is projected gradient descent on F_S.]*

2.3 The novelty operator

Lemma 1. *[Fast-student steady state gives $x_S^* = (I - N_S)x_T$, hence $\varepsilon_{TS} = -N_S x_T$, with $N_S = \pi_S S_S(\pi_{TS} I + \pi_S S_S)^{-1}$; N_S is symmetric PSD and $N_S u = 0$ on learned directions.]*

2.4 The surprise identity: the settled student's free energy is the novelty score

Corollary 1. *[At the fast-student equilibrium, $F_S^{\mathrm{eq}}(x_T) \equiv \min_{x_S} F_S(x_S, x_T) = \frac{\pi_{TS}}{2}\, x_T^\top N_S\, x_T$, hence $\nabla_{x_T} F_S^{\mathrm{eq}} = \pi_{TS} N_S x_T$: the student's surprise about the teacher's state is the novelty-weighted energy of that state.]*

2.5 The student steers the teacher up the student's surprise

Theorem 1. *[CORE: isolating the student's contribution (the push $u = |\pi_{ST}| N_S x_T$), the teacher's dynamics is exact gradient ascent on the student's settled variational free energy: $\tau_T \dot{x}_T|_{\mathrm{push}} = (|\pi_{ST}|/\pi_{TS}) \nabla_{x_T} F_S^{\mathrm{eq}}(x_T)$ (Corollary 1) — at every point, hence locally steepest ascent of the student's surprise. Consequences: (a) prioritization — axis by axis, $\dot{c}_k = (|\pi_{ST}|/\tau_T) n_k c_k$: learned ($n_k = 0$) frozen, novel amplified in proportion to novelty; (b) self-limiting — the push on a direction dies as the student learns it ($|\pi_{ST}| n_k \leq (|\pi_{ST}|\pi_S/\pi_{TS}) \|M_S u_k\|^2$); (c) the full flow is $-\nabla F_T + (|\pi_{ST}|/\pi_{TS}) \nabla F_S^{\mathrm{eq}}$: the teacher descends its own surprise while ascending the student's. Local ascent (not the global summit); the push rescales but does not seed an empty direction (noise seeds it).]*

2.6 Staying on the memory manifold

Lemma 2. *[Under the conservative guard $|\pi_{ST}| < \pi_T \sigma^2_{\min}$ the off-manifold block is strictly contracting; caveat: not exact manifold invariance (cross-block leakage measured numerically).]*

2.7 An exact free-energy saddle

Proposition 2. *[The physical functional $\Phi = F_T - F_S$ generates the sleep dynamics (on the teacher sphere, with the model's mobilities and projected W_S ascent) iff $\pi_{ST} = -\pi_{TS}$ — by Theorem 1, exactly when the teacher's ascent rate on the student's surprise matches the precision π_{TS} that defines it. The teacher descends Φ; the student, descending F_S (Proposition 1), ascends Φ: a minimax on one potential. Shape reminiscent of the active-inference agent–environment saddle, not identical.]*

2.8 Scope: subspace, not named episodes

Remark 1. *[Linearity ⇒ transfer of a subspace / effective rank, not named patterns; episodic one-per-step replay needs an added nonlinearity.]*

3 Results

3.1 Prioritized transfer of unlearned directions

[figure placeholder]
Fig 1. Novelty-spectrum "staircase" and x_T–manifold alignment: the student acquires unlearned directions; "reliable completion across tested regimes" (not global-convergence). Steps are a timescale/plotting artifact.

3.2 The wake/sleep saddle picture

[figure placeholder]
Fig 2. Saddle cross-sections: a bowl along a predictable direction vs. a hill along a novel one; the exact $\Phi = F_T - F_S$ picture at $\pi_{ST} = -\pi_{TS}$.

3.3 Merging memories by interleaving

[figure placeholder]
Fig 3. Interleaved rehearsal merges two teachers: the crosstalk sawtooth decays to zero (union of memory subspaces), demonstrated in simulation.

3.4 Continual learning without catastrophic forgetting

[figure placeholder]
Fig 4. Continual learning: retention across a streamed sequence vs. a buffer-only control that catastrophically forgets.

4 Discussion

5 Conclusion

Materials and methods

Supporting information

References

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